

BIOAVAILABILITY AMPLIFIERS: AMMONIUM COMPOUNDS & LESION PHASE CLASSIFICATION

The Role of Ammonium Compounds as Biochemical Accelerants and a Three-Phase Morphological Classification of Hamer Loop Lesions

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Abstract

This document, the second in the Hamer Loop Manuscript Series, addresses two interrelated aspects of the framework established in Document 1: the role of ammonium-containing food compounds as biochemical accelerants of Neu5Gc-mediated inflammation, and a three-phase morphological classification of the lesions produced by Hamer Loop activity.

Ammonium compounds — present in processed foods as leavening agents, colour stabilisers, pH regulators, and flavour systems — are proposed to act not merely as additional antigenic triggers but as threshold-lowering potentiators that amplify the immune response to Neu5Gc out of proportion to the antigenic load alone. Three distinct mechanisms are proposed: mucosal pH alkalinisation, direct mast cell sensitisation, and NLRP3 inflammasome priming.

The lesion phase classification introduced here represents a significant refinement to the framework: lesion characteristics — consistency, odour, depth, and distribution pattern — are proposed to be determined primarily by timing within the inflammatory cycle rather than anatomy. This temporal principle positions the Hamer Loop as the initiating phase of a process that subsequently hands off to conventional sebaceous and microbial pathophysiology, bridging the two frameworks rather than opposing them.

Keywords: *ammonium potentiation; Neu5Gc; xenosialitis; lesion morphology; TRM cells; lymphatic sequestration; sebaceous handoff; acne; inflammatory threshold; Acnerase*

1. Overview and Position Within the Series

Document 1 of this series established the foundational Hamer Loop model: dietary Neu5Gc from bovine sources is incorporated into host tissue glycans, recognised as foreign by circulating anti-Neu5Gc antibodies, and initiates a self-reinforcing inflammatory cycle

characterised by lymphatic sequestration, interstitial reservoir formation, and site-specific amplification via tissue-resident memory T cells (TRM cells).

This document addresses two important refinements. First, it formally characterises ammonium compounds as a distinct and potent class of co-trigger whose effect on the inflammatory response is qualitatively different from — not merely additive to — the Neu5Gc antigen load alone. Second, it introduces a three-phase lesion classification system based on longitudinal morphological observation, establishing that timing within the Hamer Loop cycle is the primary determinant of lesion characteristics.

Together, these two elements explain the range of lesion presentations observed in susceptible individuals and provide the observational and mechanistic foundation for the severity hierarchy documented in Appendix II (the Neu5Gc Severity Index).

2. The Ammonium Potentiation Effect

Observational data from repeated provocation trials suggest that ammonium-containing compounds do not merely add to the antigenic load. They appear to reduce the biological threshold at which the immune system mounts an inflammatory response — such that exposures that would otherwise remain subclinical instead produce visible, sometimes severe, lesions. This non-additive amplification effect is the defining characteristic of what this framework terms the Ammonium Potentiation Effect.

2.1 Observed Clinical Characteristics

The Ammonium Potentiation Effect manifests through three consistent clinical observations across repeated provocation trials:

Observation	Description
Threshold lowering	Exposures that would remain subclinical in the absence of ammonium produce visible lesions when ammonium compounds are present in the same product
Disproportionate severity	Reaction severity exceeds what the Neu5Gc load alone would predict — a small processed food exposure can trigger a response equivalent to a much larger whole-food exposure
Nodular lesion production	The highest-severity lesion presentations in the longitudinal record are consistently associated with co-exposure to Neu5Gc sources and ammonium-derived additives, producing the deep, painful nodular Category 3 presentations described in Section 3

Table 1. Clinical characteristics of the Ammonium Potentiation Effect as observed across longitudinal self-provocation trials.

2.2 Proposed Biochemical Mechanisms

The following mechanisms are proposed on the basis of mechanistic plausibility and their consistency with the observed clinical pattern. Each requires formal biochemical validation before mechanistic claims can be formalised.

2.2.1 Mucosal pH Alkalinisation

Ammonium compounds are alkaline in aqueous solution. Mucosal exposure to ammonium-bearing food ingredients may raise the local pH of the intestinal epithelium above its normal acidic range, disrupting the acid-barrier defence that partially attenuates antigen penetration. A more permeable mucosal environment allows a greater fraction of the dietary Neu5Gc present in the same food product to access subepithelial immune compartments, effectively amplifying the systemic antigenic dose beyond what the food's Neu5Gc content alone would deliver.

This mechanism also provides a plausible explanation for why liquid delivery vehicles — soups, beverages, and liquid condiments — produce accelerated Hamer Loop timelines relative to solid food sources of equivalent Neu5Gc content: the liquid matrix reduces mechanical buffering, and its ammonium-containing flavour and colour systems are delivered rapidly to the epithelial surface.

2.2.2 Direct Mast Cell Sensitisation

Ammonium compounds have been proposed to directly sensitise mast cell populations, lowering the threshold for degranulation upon subsequent antigen engagement. In the context of the Hamer Loop, this would mean that the same circulating anti-Neu5Gc antibody titre — engaging the same quantity of incorporated Neu5Gc — produces a more intense mast cell response when ammonium has been co-ingested. This threshold-lowering mechanism explains the observation of sub-2-minute Stage 1 responses (rhinorrhoea, flushing) following ingestion of ammonium-potentiated Neu5Gc sources, a timeframe inconsistent with classical IgE-mediated food allergy (typically 15–30 minutes) but consistent with pre-primed mast cell populations operating at a pharmacologically lowered activation threshold.

2.2.3 NLRP3 Inflammasome Priming

Ammonium compounds are known NLRP3 inflammasome activators. NLRP3 activation amplifies the downstream production of IL-1 β and IL-18 — cytokines that are central to the acute inflammatory response — beyond the levels generated by NF- κ B activation from Neu5Gc xenosialitis alone. The combined IL-1 β output of both pathways would be expected to produce a more intense, more rapidly escalating inflammatory response than either pathway independently. This provides the most mechanistically direct explanation for the disproportionate severity observed in ammonium-potentiated reactions.

2.3 Common Sources of Hidden Ammonium in the Food Supply

A significant practical challenge is that ammonium compounds are frequently listed under broad or non-specific label terms, making identification difficult without specialised knowledge. The following categories are the primary vectors for ammonium catalyst exposure in the food supply:

- Caramel Colour Class III and Class IV: produced by reacting carbohydrates with ammonia or ammonia-sulfite at high concentration; the most widely used food colorant globally and a Severity 5 trigger in the Neu5Gc Severity Index
- Natural flavours and artificial flavours: concentrated flavour systems delivered in solvent and reaction-flavour matrices with high ammonium-reaction-product content
- Spice and spice blends: commercial spice preparations standardised with ammoniated carriers and anti-caking agents
- Enriched flour: fortified with vitamins and iron using ammonium chloride and ammonium sulfate as dough conditioners and yeast nutrients
- Caramel flavour: caramel-based flavouring blends combining caramelised sugar with ammoniated reaction products

The Acnerase application's proprietary database of 150+ ingredient synonyms is specifically designed to surface these hidden ammonium sources — compounds that would not be identifiable through label reading alone and that transform a routine Neu5Gc exposure into an acute inflammatory event.

3. Three-Phase Lesion Classification

Longitudinal observation across repeated exposure-response cycles has identified three morphologically and temporally distinct lesion categories within the Hamer Loop framework. The governing principle of this classification is straightforward but clinically important: timing determines phase, and phase determines characteristics.

Lesion category is primarily determined by where in the Hamer Loop cycle the lesion is observed — not by its anatomical location, not by its size alone, and not by severity as conventionally measured. Anatomy is secondary. Odour is the most reliable field indicator of phase transition. This principle has remained consistent across four decades of observational data.

3.1 Category 1: Early-Phase Lesions (Lymphatic Origin)

Category 1 lesions form rapidly following dietary Neu5Gc exposure, consistent with the Stage 3 timeline described in Document 1 (2–4 hours post-exposure). They are characterised by:

- Milky, creamy consistency on expression
- No appreciable odour
- Rapid formation following trigger exposure
- Tendency toward smaller size relative to later-phase presentations
- More rapid resolution where the trigger load is not sustained

The milky consistency and absence of odour are consistent with a lymphatic-phase origin: the body is mobilising to sequester and contain the inflammatory load through lymphatic channels

before secondary microbial involvement has occurred. These lesions represent the Hamer Loop in its initial, purely immune-mediated phase.

3.2 Category 2: Later-Phase Lesions (Sebaceous-Microbial Transition)

Category 2 lesions develop more slowly, are larger, and carry a thick, cheesy consistency on expression with a pronounced gram-negative odour. These characteristics are consistent with conventional sebaceous and microbial pathophysiology — the established mechanisms of follicular occlusion and bacterial colonisation. Within the Hamer Loop model, this phase represents the handoff.

The lymphatic system, having been initially overwhelmed by the inflammatory reservoir generated in the Category 1 phase, yields to the sebaceous pathway as the body's secondary clearance mechanism. The gram-negative odour is the temporal marker: its presence signals that microbial involvement has established itself within the lesion, and that the Hamer Loop's lymphatic-dominant phase has transitioned to the conventional sebaceous-dominant phase.

This transition is the point at which the Hamer Loop model and conventional dermatological theory converge. They are not competing explanations for the same phenomenon — they are sequential phases of a single process. The Hamer Loop initiates and feeds into accepted dermatological pathophysiology rather than standing in opposition to it. This is the academically defensible framing and should be emphasised in any clinical or research context.

3.3 Category 3: Deep Cystic Lesions (Chronic Lymphatic Sequestration)

Category 3 lesions represent the most severe and clinically significant presentation within the framework. These are deep-seated, large cystic formations — colloquially described as boils — that present as lone-wolf formations without consistent anatomical clustering. Their distribution corresponds instead to the locations of deeper subcutaneous lymph node clusters, which vary in size and tissue depth across the body.

This distribution pattern is itself a compelling indicator of lymphatic involvement: the lesions appear where the lymphatic architecture concentrates, not where sebaceous activity is highest. They are most commonly associated with sustained high-fat bovine trigger exposures or ammonium-potentiated exposures, consistent with the deeper sequestration mechanisms described in this document and in Document 3.

Category 3 lesions harden over time as the body encapsulates the persistent inflammatory reservoir in fibrous tissue — a response consistent with chronic, unresolved lymphatic sequestration. Surgical expression of these lesions produces a markedly intense gram-negative odour significantly exceeding that of Category 2 lesions, consistent with a large, long-established anaerobic reservoir. Following surgical expression of the primary contents, these lesions have been observed to refill with serous, lymphatic-consistent fluid — a finding that a conventional sebaceous cyst model does not adequately explain and that represents one of the more compelling anatomical observations in support of the lymphatic sequestration hypothesis.

3.4 The Temporal Principle: A Summary

Category	Phase	Timing	Consistency	Odour	Proposed Origin	Primary Trigger Association
1	Early / Lymphatic	2–4 hrs post-exposure	Milky, creamy	None	Lymphatic sequestration of inflammatory load	Any Neu5Gc source; rapid-onset response
2	Later / Sebaceous-Microbial	Hours to days post-exposure	Thick, cheesy	Gram-negative	Handoff to sebaceous and microbial pathophysiology	Sustained exposure; higher-fat sources
3	Chronic / Deep Sequestration	Days to weeks; persistent	Hard then serous refill	Intense gram-negative	Chronic lymphatic sequestration; fibrous encapsulation	Ammonium-potentiated; prolonged high-fat exposure

Table 2. Three-Phase Lesion Classification. Timing within the Hamer Loop cycle is the primary determinant of lesion category. Odour is the most reliable field indicator of phase transition.

4. The Trigger Severity Hierarchy

Not all Neu5Gc sources produce equivalent inflammatory responses, and not all ammonium sources produce equivalent potentiation. The severity hierarchy below summarises the proposed relationship between trigger category and observed lesion characteristics, derived from longitudinal observational data across repeated exposure-elimination cycles. This hierarchy forms the basis of the Acnerase ingredient classification system and the Neu5Gc Severity Index documented in Appendix II.

Trigger Category	Severity	Delivery Mechanism	Typical Lesion Outcome
Whole-food dairy (milk, yoghurt, soft cheese)	Mild to Moderate	Portal absorption; standard antigen kinetics; no amplifiers	Category 1; small, transient; relatively rapid resolution
Concentrated dairy fats (butter, ghee, aged cheese, AMF) — bovine only	Moderate to Severe	Chylomicron-mediated lymphatic delivery; high glycan density; no chemical catalyst	Category 2; persistent; anatomically deeper
Ammonium-potentiated processed products	Severe to Suppurative	Neu5Gc load combined with threshold-lowering ammonium catalyst; NLRP3 priming	Category 2 to 3; deep nodular; prolonged resolution; greatest sequestration

Table 3. Trigger Severity Hierarchy. Fat provides antigen volume; ammonium provides inflammatory intensity. Together they define the upper range of lesion severity in the Hamer Loop framework.

The practical implication of this hierarchy is that prioritising elimination of processed derivatives, concentrated fats, and ammonium-containing products is likely to produce the greatest reduction in lesion severity and frequency, and represents the logical starting point for a staged avoidance protocol. Complete avoidance of all Neu5Gc sources — including whole milk and unprocessed red meat — is necessary to achieve full symptom resolution, but the severity hierarchy guides the most clinically impactful first steps.

5. The Dairy Fat Severity Sub-Hierarchy

Within the dairy trigger category specifically, observed lesion severity correlates with milk fat percentage. This is consistent with the role of dietary fat as a concentrated delivery vehicle for Neu5Gc: higher fat content represents a more concentrated antigen load per unit consumed, delivered via the chylomicron-mediated lymphatic route described in detail in Document 3. The sub-hierarchy below is derived from longitudinal observational data and reflects the practical severity gradient within the dairy category.

Dairy Sub-Category	Examples	Severity	Lesion Profile
Low fat (baseline)	Whole milk, skim milk, yoghurt, kefir	Mild (1)	Small, transient; relatively rapid resolution
Moderate fat	Cream, sour cream, cottage cheese	Moderate (2)	Larger; more persistent; deeper
High fat	Butter, ghee, butter oil, butter solids	Moderately Severe (3)	Severe; deeper-seated; slower resolution
Concentrated / processed fat	Anhydrous Milk Fat (AMF), aged cheese, anti-caking systems	Severe to Suppurative (4–5)	Deep nodular; acute onset; most prolonged resolution

Table 4. Dairy Fat Severity Sub-Hierarchy. Severity correlates with milk fat concentration, consistent with the chylomicron-mediated lymphatic delivery mechanism described in Document 3.

Note: Ghee, as clarified butter with the aqueous fraction removed, retains concentrated fat-soluble Neu5Gc fractions in a highly bioavailable form and produces correspondingly severe presentations. Anhydrous Milk Fat (AMF) represents the most concentrated bovine-derived fat form and appears in processed foods in quantities not intuitively apparent from label reading, making it a particularly significant Acnerase detection target.

6. Limitations

The mechanisms described in this document are proposed on the basis of longitudinal observational data and self-provocation trials conducted by a single subject. Formal biochemical validation — particularly for the mucosal pH alkalinisation, mast cell sensitisation, and NLRP3 inflammasome priming hypotheses — has not been conducted in a controlled setting.

The three-phase lesion classification is based on direct morphological observation rather than histological or biochemical analysis. Category 1 lesions have not been biopsied at the 2–4 hour window with immunofluorescence staining for IgG, IgA, or C3 deposition — the definitive test of the lymphatic-origin hypothesis described in Section 10.4 of Document 1. The proposed serous refill of Category 3 lesions following surgical expression is consistent with lymphatic fluid but has not been confirmed by fluid analysis.

The trigger severity hierarchy and dairy fat sub-hierarchy reflect observational patterns rather than controlled dose-response data. Individual variability in anti-Neu5Gc antibody titre, gut microbiome composition, TRM cell distribution, and genetic factors including HLA type and secretor status (FUT2 polymorphism) may substantially influence the severity gradient in other susceptible individuals.

These findings are offered as structured, observation-driven propositions to guide future experimental design, not as established mechanistic fact.

7. Conclusion

This document establishes two refinements to the Hamer Loop framework of central importance for understanding the clinical variability of the condition and for designing practical avoidance strategies.

The Ammonium Potentiation Effect explains why identical Neu5Gc exposures produce dramatically different outcomes depending on the food matrix in which they are delivered. Fat provides volume; ammonium provides intensity. A processed food product containing both — a common characteristic of commercial soups, sauces, confectionery, and fast food — delivers a qualitatively different inflammatory challenge than either component alone.

The three-phase lesion classification bridges the Hamer Loop model with conventional dermatology rather than competing with it. Category 1 lesions are the Hamer Loop in its purest form: rapid, lymphatic, immune-mediated, odourless. Category 2 lesions are the handoff: the point at which established sebaceous and microbial pathophysiology takes over from the immune-mediated initiating phase. Category 3 lesions are the consequence of chronic unresolved sequestration: the body's attempt to wall off what the lymphatic system could not clear.

Together, these two elements account for the full morphological range of presentations observed in susceptible individuals and provide the theoretical foundation for the severity-based ingredient classification that sits at the heart of the Acnerase platform.

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